EKREM TURAN FIRAT

Undergraduate Researcher | Ankara / Sivas, Türkiye

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I am a third-year Electrical and Electronics Engineering student and an undergraduate researcher at METU Power Lab. My main research and work area is power electronics and the design of advanced hardware for this field.

Experience

METU Power Lab — *February 2025 – Ongoing*
Undergraduate Researcher — *ODTÜ, Ankara, Türkiye*

- *Working on the design of power electronics systems such as TI C2000-controlled low-inductance motor drivers for TÜBİTAK and industrial institutions.*
- *Developing and maintaining the lab's KiCad libraries and a Claude-based KiCad design assistant.*

AGH Power Electronics Laboratory — *October 2025 – February 2026*
Undergraduate Researcher (Erasmus+) — *AGH, Kraków, Poland*

- *Worked on power electronics converter designs, such as high-speed switching with GaN FETs in synchronous buck converters.*

IEEE SCÜ Student Branch — *November 2024 – September 2025*
Committee Chairman — *Sivas, Türkiye*

- *Chairman of the PELS (Power Electronics Society) Committee.*
- *Participated in international symposiums.*

Robolink Teknoloji — *June 2023 – August 2023*
Electronics Design Intern — *OSTİM, Ankara, Türkiye*

- *Designed an ESP32-based laser CNC motherboard on a 4-layer PCB in Altium CircuitStudio, with passive power protection, USB-to-TTL and stepper motor driver circuits.*
- *Gained experience in SMD soldering, oscilloscope use and PCB testing.*

Robolink Teknoloji — *March 2022 – August 2022*
Electronics Design Intern — *OSTİM, Ankara, Türkiye*

- *Designed an independent stepper motor controller on a 2-layer PCB in EAGLE, which generates the pulse signal the driver needs on its own.*
- *Designed an ESP32 development board to verify the company's in-house design processes.*

Education

Sivas Cumhuriyet University — *September 2024 – Ongoing*
Bachelor of Science, Electrical & Electronics Engineering — *Sivas, Türkiye*

AGH University of Krakow — *October 2025 – March 2026*
Erasmus+ Exchange, Electrical Engineering — *Kraków, Poland*

Cezeri Green Technology Technical High School — *September 2020 – June 2024*
Technical Diploma, Electrical & Electronics Technology — *Ankara, Türkiye*

Projects

METU Power Lab — *ODTÜ, Ankara, Türkiye*
PowerLab KiCad Assistant — *October 2026 – Ongoing*

- *A KiCad 10 side panel that sets the rules for Claude in PCB design, so it can build an entire project to the METU Power Lab standard: requirements, schematic, libraries, placement, routing, ERC/DRC, design review and fabrication files.*

PCB Motor with Internal Driver — *October 2026 – Ongoing*

- *An axial-flux motor whose stator windings are traces on a 12-layer PCB, with the whole motor driver on a bay of the same board: three TI LMG2100 GaN half-bridges at 100 kHz (8.7 A rms, 15.4 A peak per phase), an STM32G491 MCU, linear and index Hall sensors for the absolute rotor angle, CAN FD and an isolated USB-C debugger.*
- *12 V input with reverse-polarity and hot-swap protection, plus a USB-C PD input. The rev A board (75 × 107.5 mm, 221 parts on both sides) is fully routed with a clean DRC.*

PL Driver Cycloid Board Design *July 2025 – January 2026*

- *Embedded motor driver design for a PCB motor with a cycloidal gear. As part of the TÜBİTAK 1001 project, I worked on the design of a 250 W, low-phase-inductance, TI C2000 series DSP-controlled miniature motor driver.*

PL Driver Evaluation Board Design *February 2025 – July 2025*

- *As part of the TÜBİTAK 1001 project, I participated in the hardware design of a 500 W (up to 2 kW) motor driver with extensive communication protocols and peripheral connectivity, on an advanced 8-layer PCB.*

AGH Power Electronics Laboratory *AGH, Kraków, Poland*

GaN-Based High-Frequency Switched DC-DC Synchronous Buck Converter *October 2025 – January 2026*

- *An experimental setup for laboratory classes, where students observe how changing one electrical parameter affects the others.*

Robolink Teknoloji *OSTİM, Ankara, Türkiye*

ESP32 Based Laser CNC Motherboard *June 2023 – August 2023*

- *Designed 4-layer PCBs for ESP32-based laser CNCs compatible with GRBL software.*

ESP32 Development Board *August 2022*

- *Took part in the design of a 2-layer ESP32 development board for testing ESP32-based applications within the company.*

Stepper Motor Driver *March 2022 – July 2022*

- *Designed a 2-layer PCB for stepper motor control and testing without requiring software or an external controller.*

Scholarships & Awards

TÜBİTAK *April 2025 – November 2026*

1001 Program Undergraduate Researcher Scholarship

Memberships & Organizations

IEEE Power Electronics Society (PELS) *November 2024 – Ongoing*

Student Member

Skills

PCB Design	*Altium, KiCad, EAGLE, Proteus, CircuitStudio*
3D Design	*Fusion 360, SolidWorks*
Simulation	*LTspice*

Languages

Turkish	*Mother Tongue*
English	*B2*
Polish	*A1 (Beginner)*